

Harish-Chandra Research Institute

Advanced Algebraic Geometry (Scheme Theory)**Assignment 2**

M.Sc. Mathematics, Year II

Throughout, all rings are commutative with identity. A scheme is said to be integral if it is reduced and irreducible, and normal if each of its local rings is an integrally closed domain. A morphism of schemes is said to be dominant if its image is dense.

Problem 1

Let A be a ring.

- (a) Let $e \in A$ satisfy $e^2 = e$. Prove that

$$V((e)) = D(1 - e),$$

and conclude that $V((e))$ is simultaneously open and closed in $\text{Spec } A$.

- (b) Suppose that $e_1, e_2 \in A$ are nonzero and satisfy

$$e_1^2 = e_1, \quad e_2^2 = e_2, \quad e_1 e_2 = 0, \quad e_1 + e_2 = 1.$$

Prove that $\text{Spec } A$ is the disjoint union of the nonempty open sets $D(e_1)$ and $D(e_2)$, and hence that $\text{Spec } A$ is disconnected.

- (c) Suppose conversely that

$$\text{Spec } A = Y_1 \sqcup Y_2,$$

where Y_1 and Y_2 are nonempty, closed and disjoint. Writing $Y_i = V(\mathfrak{a}_i)$, prove that

$$\mathfrak{a}_1 + \mathfrak{a}_2 = A$$

and that every element of $\mathfrak{a}_1 \mathfrak{a}_2$ is nilpotent.

- (d) Retaining the notation of part (c), choose $x_i \in \mathfrak{a}_i$ with $x_1 + x_2 = 1$, and choose an integer n such that $(x_1 x_2)^n = 0$. Expand

$$1 = (x_1 + x_2)^{2n},$$

and use the resulting decomposition to produce nonzero orthogonal idempotents $e_1, e_2 \in A$ satisfying

$$e_1 + e_2 = 1 \quad \text{and} \quad e_i \in \mathfrak{a}_i.$$

- (e) Let e_1, e_2 be nonzero orthogonal idempotents with $e_1 + e_2 = 1$. Prove that Ae_1 and Ae_2 are rings with identity elements e_1 and e_2 respectively, and that

$$A \longrightarrow Ae_1 \times Ae_2, \quad a \longmapsto (ae_1, ae_2),$$

is a ring isomorphism onto a product of two nonzero rings.

- (f) Suppose conversely that there is an isomorphism

$$A \cong A_1 \times A_2$$

with A_1 and A_2 nonzero. Find the associated pair of orthogonal idempotents of A . Combining this with the preceding parts, deduce the equivalence of the three conditions.

- (i) $\text{Spec } A$ is disconnected;
- (ii) A contains a nontrivial idempotent, equivalently there exist nonzero idempotents $e_1, e_2 \in A$ such that $e_1 e_2 = 0$ and $e_1 + e_2 = 1$;
- (iii) A is isomorphic to a product of two nonzero rings.

Problem 2

Let $f: X \rightarrow Y$ be a morphism of schemes. Recall that f is said to be *locally of finite type* if there exists an open affine cover $\{V_j = \text{Spec } B_j\}$ of Y such that, for each j , the preimage $f^{-1}(V_j)$ admits a cover by open affine subsets $\text{Spec } A_{ij}$ with each A_{ij} a finitely generated B_j -algebra.

- (a) Let $B \rightarrow A$ be a ring homomorphism making A a finitely generated B -algebra. Prove that A_a is a finitely generated B -algebra for every $a \in A$, and that

$$A \otimes_B B_g$$

is a finitely generated B_g -algebra for every $g \in B$.

- (b) Let $V = \text{Spec } B$ be an open affine subset of Y over which the finite generation condition holds, and let $V' = D(g) \subseteq V$ be a distinguished open subset. Prove that the condition holds over V' as well.
- (c) Let $W = \text{Spec } C$ be an arbitrary open affine subset of Y . Prove that every point of W has an open neighbourhood which is distinguished in W and is also distinguished in some member of the given affine cover $\{V_j\}$. Deduce that W admits a cover by such open sets.
- (d) Using part (c), prove that $f^{-1}(W)$ admits an affine open cover by spectra of finitely generated C -algebras.

Problem 3

A morphism $f: X \rightarrow Y$ of schemes is said to be *finite* if there exists an open affine cover $\{V_j = \text{Spec } B_j\}$ of Y such that, for each j , the preimage $f^{-1}(V_j)$ is affine, say

$$f^{-1}(V_j) = \text{Spec } A_j,$$

with A_j a finite B_j -module.

- (a) Prove that if A is a finite B -module and $g \in B$, then

$$A \otimes_B B_g \cong A_g$$

is a finite B_g -module. Deduce that finiteness is inherited by the preimages of distinguished open subsets.

- (b) Let M be a B -module and let $g_1, \dots, g_r \in B$ generate the unit ideal. Prove that if each localization M_{g_i} is a finite B_{g_i} -module, then M is a finite B -module.
- (c) Let $V = \text{Spec } B$ be an arbitrary open affine subset of Y . Using the distinguished-open refinement established in Problem 2(c), prove that V may be covered by finitely many distinguished open subsets $D(g_i)$ such that each

$$f^{-1}(D(g_i))$$

is affine with coordinate ring finite over B_{g_i} . Prove moreover that $g_1, \dots, g_r$ generate the unit ideal of B .

- (d) Using the affine criterion from Assignment 1, Problem 5(b), prove that $f^{-1}(V)$ is affine.
- (e) Writing

$$f^{-1}(V) = \text{Spec } A,$$

prove that A is a finite B -module, and deduce the asserted equivalence.

Problem 4

Let X be an integral scheme with function field K . For an open affine subset $U = \operatorname{Spec} A \subseteq X$, let $\tilde{A}$ denote the integral closure of A in $\operatorname{Frac}(A)$, and set

$$\tilde{U} = \operatorname{Spec} \tilde{A}.$$

- (a) Prove that $\operatorname{Frac}(A) = K$ for every nonempty open affine subset $U = \operatorname{Spec} A$ of X , and that $\tilde{A}$ is an integrally closed domain with fraction field K . Deduce that $\tilde{U}$ is a normal integral scheme.
- (b) Prove that integral closure commutes with localization, in the sense that for $0 \neq a \in A$ the integral closure of A_a in K is $\tilde{A}_a$. Deduce that the normalization of a distinguished open subset of U is the corresponding distinguished open subset of $\tilde{U}$.
- (c) Let U and U' be open affine subsets of X . Using part (b), construct a canonical isomorphism between the inverse images of $U \cap U'$ under

$$\tilde{U} \longrightarrow U \subseteq X \quad \text{and} \quad \tilde{U}' \longrightarrow U' \subseteq X.$$

Verify the cocycle condition on triple overlaps.

- (d) Deduce that the schemes $\tilde{U}$ glue to a scheme $\tilde{X}$ equipped with a morphism

$$\nu: \tilde{X} \rightarrow X,$$

and prove that $\tilde{X}$ is integral and normal.

- (e) Let Z be an integral scheme with function field $K(Z)$, and let $W \subseteq Z$ be a nonempty open subset. Prove that

$$\mathcal{O}_Z(W) = \bigcap_{z \in W} \mathcal{O}_{Z,z}$$

inside $K(Z)$. Deduce that if Z is normal, then $\mathcal{O}_Z(W)$ is integrally closed in $K(Z)$.

- (f) Let Z be a normal integral scheme and let

$$g: Z \rightarrow X$$

be a dominant morphism. Prove that g induces an injection

$$K \hookrightarrow K(Z)$$

of function fields. For $U = \operatorname{Spec} A \subseteq X$ open affine and $W = g^{-1}(U)$, prove that the homomorphism

$$A \rightarrow \mathcal{O}_Z(W)$$

extends uniquely to a homomorphism

$$\tilde{A} \rightarrow \mathcal{O}_Z(W).$$

Show that the resulting morphisms $W \rightarrow \tilde{U}$ glue to a morphism

$$h: Z \rightarrow \tilde{X}$$

satisfying $\nu \circ h = g$, and prove that h is unique.

- (g) Assume that X is of finite type over a field k . Using the finiteness theorem for the integral closure of a finitely generated k -domain in its fraction field, prove that

$$\nu: \tilde{X} \rightarrow X$$

is a finite morphism.

Problem 5

For products of classical varieties, the Zariski topology need not be the product topology. For schemes, there is an additional observation: even the underlying set of a fibre product need not be the Cartesian product of the underlying sets.

- (a) Let k be a field. Prove that

$$k[x] \otimes_k k[y] \cong k[x, y]$$

as k -algebras, and deduce from the universal property of the fibre product that

$$\mathbb{A}_k^1 \times_{\operatorname{Spec} k} \mathbb{A}_k^1 = \mathbb{A}_k^2.$$

- (b) Describe the two projection morphisms

$$\mathbb{A}_k^2 \rightarrow \mathbb{A}_k^1$$

on points, in terms of prime ideals.

- (c) Assume that k is algebraically closed, and let

$$\pi: |\mathbb{A}_k^2| \longrightarrow |\mathbb{A}_k^1| \times |\mathbb{A}_k^1|$$

denote the map of underlying sets induced by the two projections. Prove that π is surjective and restricts to a bijection on closed points, but that π is not injective. Determine the fibre of π over the pair consisting of the generic point of each factor, and verify that this fibre is infinite.

- (d) Let s and t be indeterminates over a field k , so that $\operatorname{Spec} k(s)$, $\operatorname{Spec} k(t)$ and $\operatorname{Spec} k$ each consist of a single point. Prove that

$$k(s) \otimes_k k(t)$$

is the localization of $k[s, t]$ at the multiplicative set generated by the nonzero polynomials in s alone together with the nonzero polynomials in t alone.

- (e) Using part (d), describe the points of

$$\operatorname{Spec} k(s) \times_{\operatorname{Spec} k} \operatorname{Spec} k(t)$$

in terms of prime ideals of $k[s, t]$. Determine the dimension of this scheme, and determine whether its underlying topological space consists of a single point.

Problem 6

Throughout this problem, closed subschemes are understood up to the usual equivalence of closed immersions.

- (a) Let $A \rightarrow A/\mathfrak{a}$ be a surjection of rings and let $A \rightarrow A'$ be an arbitrary ring homomorphism. Prove that

$$A' \rightarrow A' \otimes_A A/\mathfrak{a}$$

is surjective, and identify its kernel.

- (b) Deduce that closed immersions are stable under base extension: if $Y \rightarrow X$ is a closed immersion and $X' \rightarrow X$ is any morphism, then

$$Y \times_X X' \rightarrow X'$$

is a closed immersion. Justify the reduction to the affine case.

- (c) Let Y be a closed subscheme of $X = \operatorname{Spec} A$. Prove that every point of Y possesses an open neighbourhood in Y of the form

$$D(h) \cap Y$$

with $h \in A$, and that such a neighbourhood is affine.

- (d) Prove that finitely many such elements h suffice to cover Y , and that, after adjoining further elements h with

$$D(h) \cap Y = \emptyset,$$

one may assume that the corresponding sets $D(h)$ cover X . Conclude that these elements generate the unit ideal of A .

- (e) Using the affine criterion from Assignment 1, Problem 5(b), prove that Y is affine.
 (f) Writing $Y = \operatorname{Spec} C$, prove that

$$A \rightarrow C$$

is surjective, and conclude that Y is the closed subscheme determined by an ideal $\mathfrak{a} \subseteq A$.

- (g) Let $Y \subseteq X$ be a closed subset endowed with the reduced induced structure, and let Y' be any closed subscheme of X with the same underlying topological space. Prove that the closed immersion

$$Y \rightarrow X$$

factors through Y' , so that the reduced induced structure is the smallest subscheme structure on Y .

- (h) Let $g: Z \rightarrow X$ be a quasi-compact and quasi-separated morphism. Prove that

$$\ker(\mathcal{O}_X \rightarrow g_*\mathcal{O}_Z)$$

is a quasi-coherent sheaf of ideals; that the closed subscheme Y which it determines is one through which g factors; and that $Y \rightarrow X$ factors through every closed subscheme of X through which g factors. Prove that Y is the unique closed subscheme with this property. It is called the *scheme-theoretic image* of g .

- (i) Prove that if Z is reduced, then the scheme-theoretic image Y of part (h) carries the reduced induced structure on

$$\overline{g(Z)}.$$

Problem 7

Let X be a noetherian topological space, and let P be a property of closed subsets of X . Assume that for every closed subset $Y \subseteq X$, if every proper closed subset of Y has property P , then Y has property P .

- (a) Prove that X satisfies the descending chain condition on closed subsets, and deduce that every nonempty family of closed subsets of X contains a minimal member.
 (b) Prove that the inductive hypothesis forces the empty set to possess P .
 (c) Suppose that X fails to possess P . Prove that the family of closed subsets failing to possess P is nonempty, select a minimal member Y of this family, and derive a contradiction.
 (d) Conclude that P holds for X .